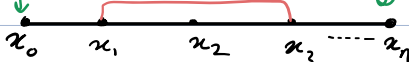
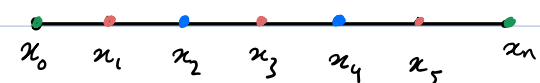


Composite Numerical integration

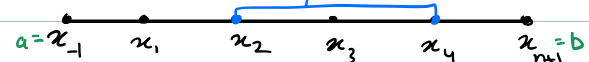
① Composite Trapezoidal rule

$$\int_{a=x_0}^{b=x_n} f(x) dx = \frac{h}{2} \left[f(x_0) + 2 \sum_{j=1}^{n-1} f(x_j) + f(x_n) \right], \quad h = \frac{b-a}{n}$$


② Composite Simpson's rule n = must be even

$$\int_{a=x_0}^{b=x_n} f(x) dx = \frac{h}{3} \left[f(x_0) + 2 \sum_{j=1}^{\text{even point}} f(x_{2j}) + 4 \sum_{j=1}^{\text{odd point}} f(x_{2j-1}) + f(x_n) \right], \quad h = \frac{b-a}{n}$$


③ Composite Midpoint Rule "Always start from x_{-1} "

$$\int_{a=x_{-1}}^{b=x_{n+1}} f(x) dx = 2h \left[\sum_{j=1}^n f(x_j) \right], \quad h = \frac{b-a}{n+2}$$


Q10

1. Use the Composite Trapezoidal rule with the indicated values of n to approximate the following integrals.

a. $\int_1^2 x \ln x \, dx, \quad n = 4$ $h = \frac{b-a}{n} = \frac{2-1}{4} = \frac{1}{4} = 0.25$

$$\begin{aligned} \int_{x_0}^{x_4} f(x) \, dx &= \frac{h}{2} [f(x_0) + 2(f(x_1) + f(x_2) + f(x_3)) + f(x_4)] \\ &= \frac{0.25}{2} [0 + 4.546842925 + 3.295836866] \\ &= 0.9806514481 \end{aligned}$$

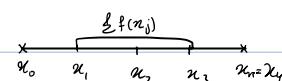
$$x_0 = 1$$

$$x_1 = a + jh = 1.25$$

$$x_2 = 1 + 2 \times 1.25$$

$$x_3 = 2$$

Stop $\rightarrow x_4 = 3$
at $n=4$



e. $\int_0^2 e^{2x} \sin 3x \, dx, \quad n = 8$

$$h = \frac{b-a}{n} = \frac{2-0}{8} = 0.25 \quad x_j = a + jh$$

$$\begin{aligned} \int_0^2 e^{2x} \sin 3x \, dx &= \frac{0.25}{2} [f(x_0) + 2[f(x_1) + f(x_2) + f(x_3) + f(x_4) \\ &\quad + f(x_5) + f(x_6) + f(x_7)] + f(x_8)] \\ &= 13.57597939 \end{aligned}$$

$$x_0 = 0$$

$$x_1 = 0 + 1 \times 0.25 = 0.25$$

$$x_2 = 0.5$$

$$x_3 = 0.75$$

$$x_4 = 1$$

$$x_5 = 1.25$$

$$x_6 = 1.5$$

$$x_7 = 1.75$$

Stop $\leftarrow x_8 = 2$
at $n=8$

Q30 3. Use the Composite Simpson's rule to approximate the integrals in Exercise 1.

$$g. \int_3^5 \frac{1}{\sqrt{x^2-4}} dx, \quad n=8 \quad \int_{x_0}^{x_n} f(x) dx = \frac{h}{3} [f(x_0) + 2 \sum_{\text{even}} f(x_{2j}) + 4 \sum_{\text{odd}} f(x_{2i-1}) + f(x_n)]$$

$$h = \frac{5-3}{8} = 0.25$$

$$\int_{x_0=3}^{x_8=5} \frac{1}{\sqrt{x^2-4}} dx = \frac{0.25}{3} [f(x_0) + 2 \sum_{\text{even}} f(x_{2j}) + 4 \sum_{\text{odd}} f(x_{2i-1}) + f(x_8)]$$

$$= 0.6043940$$

$$\begin{aligned} x_0 &= 3 \\ x_1 &= 3 + 1 \times 0.25 = 3.25 \text{ odd} \\ x_2 &= 3.5 \text{ even} \\ x_3 &= 3.75 \text{ odd} \\ x_4 &= 4 \text{ even} \\ x_5 &= 4.25 \text{ odd} \\ x_6 &= 4.5 \text{ even} \\ x_7 &= 4.75 \text{ odd} \\ x_8 &= 5 \end{aligned}$$

Stop at $n=8$
 x_n

Q50 5. Use the Composite Midpoint rule with $n+2$ subintervals to approximate the integrals in Exercise 1.

$$d. \int_0^{\pi} x^2 \cos x dx, \quad n=6 \quad h = \frac{b-a}{n+2} = \frac{\pi-0}{6+2} = \frac{\pi}{8}$$

$$\int_0^{\pi} x^2 \cos x dx = 2h \sum_{\text{even point}} f(x_{2j})$$

$$= 2 \frac{\pi}{8} [-3.4894321]$$

$$= -2.7405935$$

$$x_{-1} = 0$$

$$x_1 = a + 1 \times h = \frac{\pi}{8}$$

$$\rightarrow x_2 = \frac{\pi}{4}$$

$$x_3 = \frac{3\pi}{8}$$

$$\rightarrow x_4 = \frac{\pi}{2}$$

$$x_5 = \frac{5\pi}{8}$$

$$\rightarrow x_6 = \frac{3\pi}{4}$$

$$x_{n+1} = x_7 = \frac{7\pi}{4}$$